YASHASHWINI AWATE

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EXPERIENCE

Research Assistant and Tutor Human-Centered AI · Otto von Guericke University *Magdeburg, Germany | April 2026 - Present*

Guiding students to build AI systems that are ethical, explainable, and designed around real human needs.

Tech Stack Used - Python, NLP, LLM, Transformers, SHAP/LIME, LangChain, Jupyter, Gen AI

Full Stack Developer Analyst | Software ML Developer · TIAA *Pune, India | June 2023 - March 2025*

Built and deployed end-to-end ML pipelines and analytics tools that improved data efficiency by 35% and cut post-release issues by 25%.

Tech Stack Used - Python, Pyspark, Snowflake, AWS (S3, EMR, Athena), Power BI, Tableau, SQL, Docker, Azure ML

Machine Learning Research Intern · Redgate's IT Solutions *Remote, India | June 2022 - June 2023*

Turned raw, messy data into actionable business insights through predictive models and interactive dashboards.

Tech Stack Used - Python, Scikit-learn, TensorFlow, Sentence-BERT, DBSCAN, Pandas, Matplotlib

Data Analyst Intern · S B Awate Consultancy *Solapur, India | March 2020 - April 2022*

Analyzed ESIC data across factories and schools to build dashboards that drove smarter compliance and business decisions.

Tech Stack Used – Advance M.S Excel, SQL, Power BI, Data Visualization

EDUCATION

Master of Science - Data and Knowledge Engineering (AI/ML Specialization) *March 2025 - Present (Expected 2027)*

Otto von Guericke University Magdeburg, Germany.

Bachelor of Technology - Computer Science and Engineering *Solapur, India · 2019 - 2023*

Dr. Babasaheb Ambedkar Technological University, First Class with Distinction

KEY PROJECTS

Understanding Memorization and Recall in Generative Models (Advanced Topics In Deep Learning) *Research Project, February 2026*

Studying how generative models memorize training data - with implications for privacy, safety, and model evaluation.

NYC Citi Bike Operational Intelligence (Data Science with Python) *January 2026*

Engineered an AI-driven bike-share logistics system on 4.7M NYC trips - covering data pipeline design, demand forecasting, and station-level inventory optimization.

Investigating Response-Driven Prompting Methods for LLM Efficiency (Recent Topics In Business Informatics) *Research paper, November 2025*

Researching how response-driven prompting strategies reduce token usage and improve output quality in large language models.

AI Risk Assessment Analysis System (Human Centered Artificial Intelligence) *August 2025*

Built an ML-NLP pipeline using Sentence-BERT and DBSCAN for contextual AI risk detection and mitigation, with interpretable outputs for non-technical stakeholders.

Columnar Data Migration and Optimization (Software Machine Learning Developer, TIAA) *January 2025*

Migrated large-scale columnar datasets from Snowflake to ODS, improving query performance and downstream analytics efficiency.

Automated ML Pipelines for Data Ingestion (Full Stack Developer, Analyst, TIAA) *June 2024*

Developed a scalable, production-ready pipeline for ingesting and preprocessing multi-source data using Python and AWS.

Interview Prep AI Platform (Bachelor of Technology, Final year project) *Final Year 2023*

NLP-powered chatbot simulating real interview scenarios with personalized preparation insights.

TECHNICAL SKILLS

Programming and ML: Python, SQL, PyTorch, Pyspark, TensorFlow, Scikit-Learn, Pandas, NumPy, Seaborn

AI and NLP: LLM Prompting, Lang Chain, Llama Index, Transformers, Sentence-BERT, Explainable AI, Responsible AI, Model Interpretability, LLM

Data and Analytics: Power BI, Tableau, Streamlit, EDA, Feature Engineering, Data Visualization, Google Big Query

Cloud and DevOps: AWS (S3, EMR, Athena, QuickSight), Azure ML, Snowflake, Docker, CI/CD, Git

Tools: MS Excel (Advanced), Microsoft SSIS, Jupyter Notebook, Agile - Scrum, Figma, Canva

ACHIEVEMENTS, CERTIFICATIONS And LEADERSHIP

HCAI Tutor, *Otto von Guericke University Magdeburg, mentoring students in ethical and explainable AI design* *April 2026-Present*

Accepted into **McKinsey.org Forward Program** *selective global leadership and skills accelerator by McKinsey.org* *March 2026-Present*

Microsoft Generative AI *for* **Data Analysis Professional Certified**, *Microsoft Power BI Data Visualization Certified* *August 2025*

AWS Certified Machine Learning, **Google Data Analytics Professional Certified**, *Hacker Rank Gold Python and Statistics* *2023-2025*

Best Outgoing Student *of University and CSE Department - Dr. Babasaheb Ambedkar Technological University* *Awarded 2023*

Training and Placement Coordinator · *NK Orchid College Of Engineering and Technology* *June 2023*

Event Coordinator and Student Council Leader · *IEI - Institution of Engineers India* *Awarded 2021-2023*

Welfare Committee Member and Social Outreach Volunteer *2019-2023*

Languages - English C1 (Certified Fluent), German A1 and A2 (Certified), currently advancing to B-level

Interests - Creative Designer, Travelling, Fiction Movies, Content Creation, Reading Philosophy, Mentoring, AI Tool Exploration.